

ENPH 453 Advanced Physics Laboratory
E2 Electron Spin Resonance

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Executive Summary

The purpose of this experiment was to identify the hyperfine resonance for DPPH and the ion impurities in the MgO crystal (Fe^{3+}, Cr^{3+}, Mn^{2+}). The ability to observe such phenomena comes from Electron Spin Resonance (ESR) which measures the absorption of microwave radiation with unpaired electrons in a magnetic field [1]. When the microwave photon energy matches the energy of separation between the spin states ($h\nu = g\mu_B H_0$) through the Zeeman interaction, resonance occurs [2]. The further interaction of the unpaired electron with a neighboring nuclear-dipole causes hyperfine splitting from the splitting of the resonance [3]. This experiment aims to provide an understanding of such occurrences by observing the hyperfine resonance in the MgO sample.

The MgO crystal placed inside the microwave resonant cavity is supplied with a fixed frequency of 200 Hz. A static magnetic field is generated by an electromagnet powered by a linear ramp supply that allows for linear sweeps to be performed and a reading to be observed with the Gaussmeter. The Gaussmeter was initially tuned by locating the Lissajous, reading the corresponding reading for H that coincided with the DPPH, and comparing it with the expected value. Linear sweeps were performed for 2 weeks with the microwave oriented at 0 ° and then 30°. The collected data for both orientations is shown in the figure below.

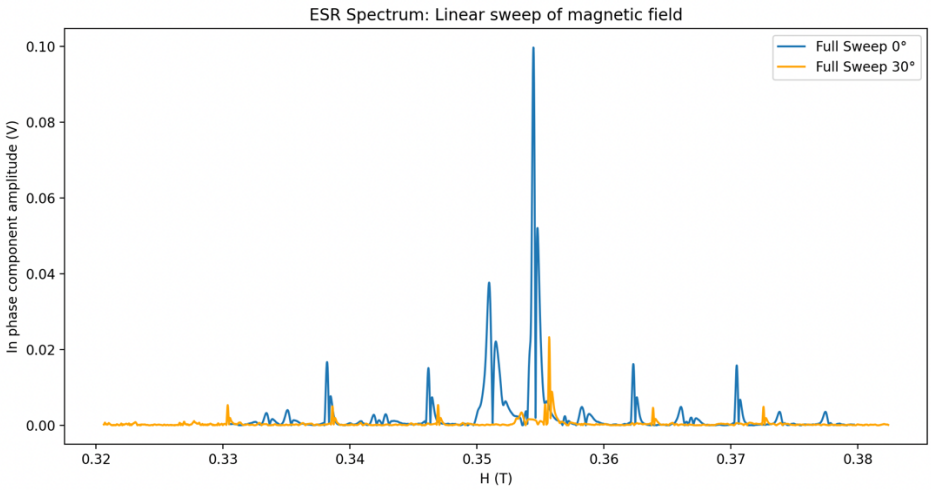


Figure: A plot of the full sweep of the full in phase absolute amplitude against the magnetic field for both 0° (blue) and 30° (orange).

After conducting these sweeps, DPPH was confidently identified and some of the ion impurities were speculated to have been identified. However, by increasing the range and more tuning potentially all ion impurities and resonance lines could have been identified.

Abstract

The purpose of this experiment was to investigate the Electron Spin Resonance (ESR) of an MgO sample. The Magnesium Oxide crystal contains transition metal impurities including Fe^{3+}, Cr^{3+}, Mn^{2+} with further experimentation yielding multiple resonance lines within the ESR spectra as a result of the interactions between the magnetic ions. The MgO crystal was placed in a microwave resonant cavity with the use of Diphenyl Picryl Hidrazyl (DPPH) as a calibration source. A linear sweep was performed and the Gaussmeter, placed near the cavity, displayed the magnetic field strength of the generated static magnetic field. From the Lissajous that was displayed on the digital oscilloscope, an experimental value for DPPH was recorded (H=0.3450 T) and compared to the expected value (H=0.2546 T). This was used to set the range for the sweep to locate the DPPH resonance peak. Data was collected over 2 weeks by performing various sweeps and collecting data at various microwave orientations. The intention of the varying sweeps would be to render the resonance peaks for the ion impurities visible. Data was collected for 0° and 30° to attempt to identify all hyperfine resonances. The data was plotted for each orientation, integrated, and then compared to the expected values for the magnetic field for each peak.

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1 Introduction

Electron paramagnetic resonance (EPR) and electron spin resonance (ESR) is a spectroscopic method first invented by Russian physicist in 1944 before being further expanded upon by a group of physicists from Oxford University [4]. It is the interaction of unpaired electron(s) in a magnetic field during the absorption of microwave radiation [1]. These unpaired electrons, when inside a magnetic field, have their degeneracy lifted as described by Eq. 1

$$\widehat{H}_s = g\mu_B B\widehat{S_z} \tag{1}$$

where the g-value is g=2.00232 for a free electron, $\widehat{H}_s$ is the spin Hamiltonian, B is the magnetic field, and μ_B is the Bohr magneton (9.274×10^{-28} JG^{-1}) [4].

The energy differences between two levels that are predominately studied in ESR are from the interactions of unpaired electrons [5]. This is known as the Zeeman effect where the energy level in the magnetic field splits [4]. When the two spin states or ($m_s = \pm\frac{1}{2}$) split, it produces an energy of separation between the two levels given by energy [6].

$$\Delta E = g\mu_B H_0 \tag{2}$$

When the energy of the microwave photons, $h\nu$, equals this energy, resonance occurs and is now given by [6].

$$h\nu = g\mu_B H_0 \tag{3}$$

This process is able to provide a quantum mechanical picture of the occurrence of resonance and the implications of the splitting of energy levels. The experiment will further an understanding of these processes through employing ESR on an MgO crystal sample.

2 Theory

The interaction of electrons in a static magnetic field (H) with a magnetic moment described in (Eq. 4)

$$\vec{\mu} = g\mu_B S \tag{4}$$

where μ_B is the Bohr magneton, g is the electron g-factor, and S is the electron spin [7]. This forms the basis of electron spin resonance [7]. An electron isolated without any outside forces still has an intrinsic angular momentum or spin (S) assigned [5]. This angular motion from the charged electron generates a magnetic field that causes it to act like a magnetic dipole with the previously affirmed magnetic moment $\vec{\mu}$ [5]. The interaction of the unpaired electron spin moments defined as $m_s = \pm 1/2$ with a magnetic field, known as the Zeeman effect, cause the spin to gain different energies defined as (Eq. 5) [4].

$$E_{m_s} = gm_B S \tag{5}$$

Another perturbation of the electron energy or another term in the spin Hamiltonian is added after an unpaired electron interacts with additional magnetic nucelei resulting in (Eq. 6) [4].

$$\widehat{H}_s = g\mu_B B\widehat{S_z} + A_0\vec{I}\cdot\vec{S} \tag{6}$$

The multiplying factor for $\vec{I}\cdot\vec{S}$ is given by A, the theoretical isotropic hyperfine coupling constant (Eq. 7)

$$A_0 = \frac{2\mu_0}{3} g\beta_e\beta_n|\psi(0)|^2 \tag{7}$$

and the hyperfine splitting constant is given $a_0 = A_0/\, g\beta_e$ [3][8]. The interaction of the unpaired electron with a neighboring nuclear-dipole moment is called hyperfine interaction or hyperfine splitting due to the resulting splitting of the resonance [3]. This interaction may be isotropic (orientation-independent) or anisotropic (orientation-dependent) [3]. The interactions between these magnetic nuclei and the electron spin are known as hyperfine coupling [9].

For nuclei with spin I, the electron spin levels split into 2I+1 equally spaced sublevels where $m_I = I, I-1, \dots 0, \dots, -I+1, -I$ is allowed for magnetic dipolar transitions [5][3]. The eigenfunctions of the spin Hamiltonian (Eq. 6) can be expressed in terms of $|m_s, m_I\rangle$ where m_I and m_S are the electron and nuclear spin quantum numbers, respectively [3]. The energy eigenvalues for the spin Hamiltonian are now expressed as (Eq. 7-10)

$$E(\frac{1}{2},\frac{1}{2}) = \frac{1}{2}g\mu_B B + \frac{1}{4}A \tag{8}$$

$$E\left(-\frac{1}{2},-\frac{1}{2}\right)=-\frac{1}{2}g\mu_B B+\frac{1}{4}A \tag{9}$$

$$E_+=\frac{1}{2}g\mu_B B\left[1+\left(\frac{A}{g\mu_B B}\right)^2\right]^{\frac{1}{2}}-\frac{1}{4}A \tag{10}$$

$$E_-=-\frac{1}{2}g\mu_B B\left[1+\left(\frac{A}{g\mu_B B}\right)^2\right]^{\frac{1}{2}}-\frac{1}{4}A \tag{11}$$

where if the hyperfine coupling is sufficiently small, the diagonal elements of the Hamiltonian matrix will be sufficiently accurate and become (Eq. 12) [3].

$$E=\pm\frac{1}{2}g\mu_B B\pm\frac{1}{2}A \tag{12}$$

The hyperfine interaction can be observed in the ESR spectra where the splitting caused by placing a sample in a uniform magnetic field also gives rise to a g-factor.

3 Apparatus and Procedure

3.1 Apparatus

A complete breakdown of the apparatus that was used in the experiment is outlined below. This is then followed by the procedure and how each piece of equipment was used and tuned for each step of the process. All components with their respective labels can be found in Figure 1 from the following breakdown:

1. National Instruments TDS 2022C 200MHz Digital Storage Oscilloscope
2. Stanford Research Systems SR810 DSP Lock-in Amplifier
3. Agilent 33220A Function Generator
4. Bryston Audio Amplifier
5. Varian Reflex Klystron
6. Hewlett Packard Klystron Power Supply
7. Linear Ramp DC Voltage Supply and KEPCO bipolar operational power supply/amplifier
8. MgO crystal and DPPH calibration sample
9. Gaussmeter
10. LabVIEW Interface

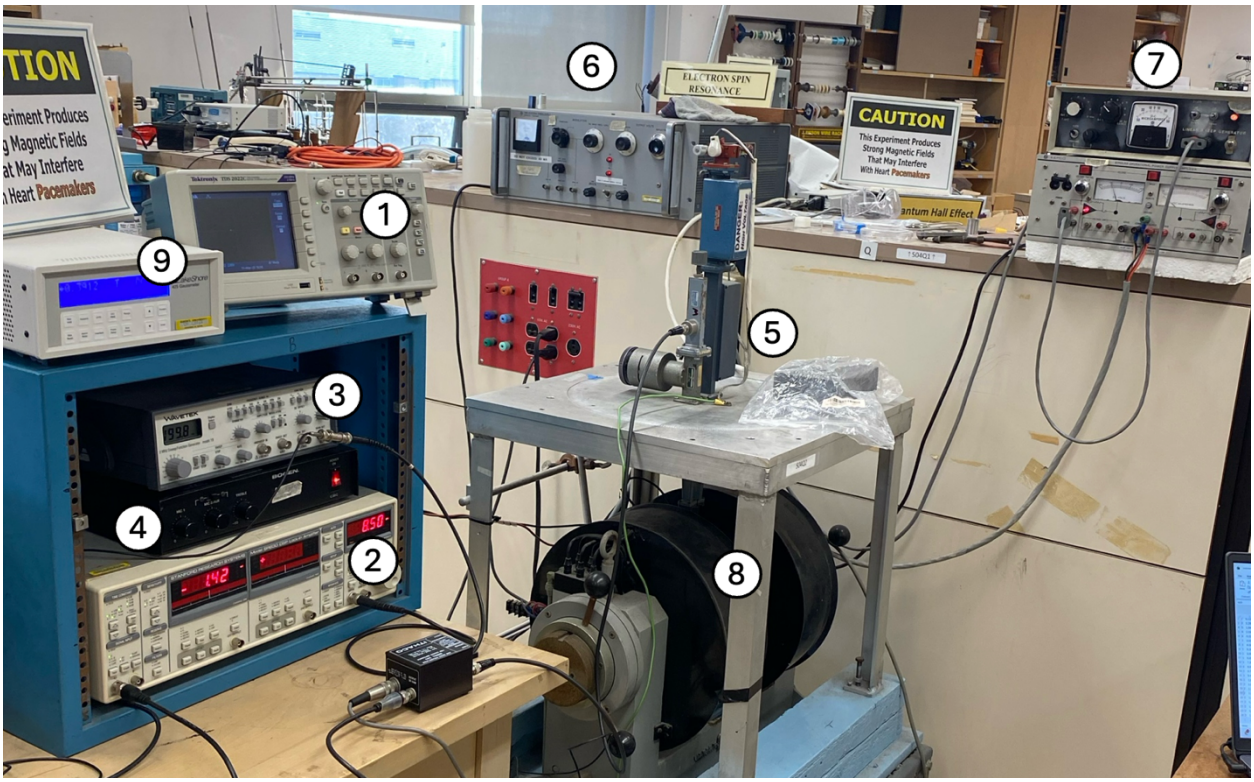


Figure 1: Labelled image of apparatus for the Electron Spin Resonance experiment. This does not include the LabVIEW interface.

The Varian Reflex Klystron is coupled to and powered by the Hewlett Packard Klystron Power Supply that provides a high voltage supply to the Klystron. It is always ensured that the current was not set to a value that would exceed 40mA. A single electromagnet provides the basis for the electromagnetic field with the MgO crystal while another magnet creates another field at a much lower intensity. The MgO crystal is located inside the Klystron in a microwave cavity while a fixed frequency is supplied by the function generator amplified by the audio amplifier. The electromagnets are supplied power by the linear ramp DC voltage supply where the intensity and resonance of the field can vary depending on the current that is supplied [10]. This generates a static magnetic field that is consistently monitored and calibrated by the Gaussmeter to find the calibration value for the DPPH source.

The resonance is viewed on the digital oscilloscope and detected by monitoring the intensity of the microwave radiation. The ESR absorption is detected using the X-Y mode of the oscilloscope and using a lock-in amplifier set to an optimal amplitude. The lock-in amplifier and Gaussmeter are connected to LabVIEW and a magnetic field is recorded alongside the X- and Y-signals as voltages. After conducting preliminary sweeps with the linear voltage ramp and locating areas of interest, multiple sweeps are conducted at 0°, 30°, and 90° to identify the hyperfine resonances of Fe^{3+}, Cr^{3+}, and Mn^{2} using the different microwave orientations.

3.2 Procedure

The experimental procedure is outlined in the following steps:

1. The microwave power supply was turned on by first flipping the switch to “FIL”, waiting 2 minutes for the microwave to heat up on, then turning it completely on.
2. The dial of the cavity was rotated off of its minimum to an arbitrary frequency and the microwave output from the Klystron power supply was connected to the oscilloscope. The dial was then set to its minimum where it couples with the magnetic field of the microwave and recorded the frequency.
3. The modulating (secondary) magnetic field was connected to X input (channel 1) and the cavity output was connected to Y input (channel 2) of the oscilloscope. The oscilloscope was set to XY plotting. The Klystron was set to the CW function and the function generator was set to 200 Hz sine wave before beginning to sweep H and tune for the modulated magnetic field at resonance. This should produce the Lissajous curve. The observed H value at this point is used to calculate the expected H_0 value for the DPPH.
4. The output of the cavity was then connected to the input of the lock-in amplifier and the output of the Audio amplifier to be the reference signal of the lock-in amplifier. Preliminary sweeps were conducted using the linear ramp supply in order to gauge any regions of interest.
5. The time constant and sensitivity of the lock-in amplifier was continually tuned until the lock-in amplifier was stabilized. The optimal phase of the lock-in amplifier was also tuned and found (-59.9°).
6. The linear ramp supply was set to an optimal level for sweeping (Time at 8 and Range at 4). This was done by adjusting H value using the gaussmeter with the FAST switch. The Bogen audio amplifier and function generator amplitude were set low (around 1-2). A sweep was started by flipping the SLOW switch up and the LabVIEW program was started. After the sweep from approximately 0.3500 to 0.3800 T the data was exported plotted, and the data was interpolated. The highest peak during the sweep was found to be DPPH.
7. The sensitivity and time constant were continually adjusted before performing additional sweeps in order to reduce the noise and provide a clearer spectrum.
8. The angle was changed to 90° with a higher sensitivity before performing a sweep to find the ion impurity peeks. Next the angle was changed to 30° to sweep for the additional ion impurities.

3.2 Procedure Errors

Errors were continually made during the course of the experiment. The first error was a broken cable that was initially connected from the function generator to the lock-in amplifier. Another more significant error was that the audio amplifier was set to the maximum amplitude. This caused the data and readings to be mostly noise, and all sweeps had to be redone with the audio amplifier set low. A low setting is required for the audio amplifier since a high amplitude would increase the strength of the secondary magnetic field and force the cavity output to a high resonance signal. This causes the hyperfine interactions to be indiscernible from noise.

It was also later discovered that the Klystron was incorrectly attenuated causing the lock-in amplifier to be continuously overloaded when the angle orientation would be changed. Therefore, whenever the angle was changed, the Klystron would have to be tapped in order to stabilize the frequency and create a clear Lissajous once more.

3.4 Safety Procedures

A main instrument in this lab is the the Hewlett Packard Klystron Power Supply that provides a high voltage to the Klystron. Due to the dangers of working with high voltages, it was always ensured that the power supply was turned on first before waiting two minutes and then turning on the high voltage power supply. This ensures that the power supply was sufficiently turned on and then also turned off to avoid overheating and causing damage to any electronics. All electrical components were always checked to be correctly turned off before leaving the lab.

During the lab, all equipment and instruments were always handled with care, and all necessary precautions were followed. This included not touching any wires on top of the Klystron as they contain high voltage coming from the power supply. The crystal and interior of the resonator cavity was never handled or obstructed while current was flowing through the electromagnet. Due to the dangers of the high voltage components and strong magnetic fields, individuals with pacemakers or heart medical devices were also advised to avoid the apparatus while the experiment was running to avoid any interference or damage medical devices.

4 Data Collection and Analysis

This section details the means of data collection and the subsequent analysis for the data that was collected. It contains the final calibration settings for the experimental apparatus, initial calculations, and the raw data for trying to find the electron spin resonance.

4.1 Calibration

During the experiment, the apparatus that was outlines above was calibrated with the settings outlined in **Error! Reference source not found.**

Table 1: Calibration settings that were tuned and adjusted throughout the experiment. These settings were used for the final data collection.

Instrument	Setting(s)
Stanford Research Systems SR810 DSP Lock-in Amplifier	Time constant = 3s Sensitivity = 100mV
Agilent 33220A Function Generator	Frequency = 200 Hz
Bryston Audio Amplifier	Amplitude 1-2
Hewlett Packard Klystron Power Supply	Continuous-Wave Mode
Linear Ramp DC Voltage Supply	Time interval = 8 Range = 7
LabVIEW Interface	Period = 500ms

The initial calibration for DPPH was done using Eq. 3 to calculate the expected value of the magnetic field using the known value g-factor of DPPH g=2.0036.

$$H_{\mathrm{DPPH}} = \frac{\mathrm{h}\nu}{g\mu_B} = \frac{(6.626x10^{-34}\ Js)(9.945x10^{9}\ Hz)}{(2.0036)\left(9.2741 \times 10^{-24}\ \frac{A}{m^2}\right)} = 0.3546\ T$$

From tuning and finding the Lissajous, the experimental value of H was found to be H=0.345 Tesla.

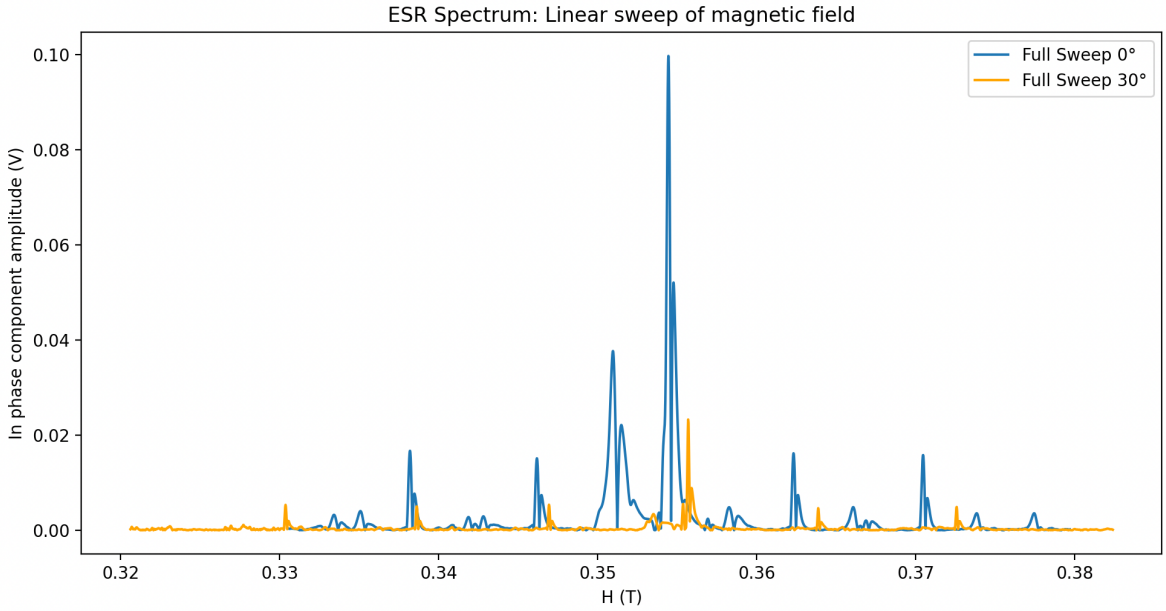


Figure 2: A plot of the full sweep of the full in phase absolute amplitude against the magnetic field for both 0° (blue) and 30° (orange).

The ESR spectra for the full linear sweep at 0° and 30° are plotted in Figure 2 showing the interpolated and shifted H field against the absolute values for the amplitude.

The ESR data collected is the derivative of the absorption and to access the original absorption spectrum the data has to be integrated. The data is presented as a first derivative initially due to the absorption of the microwave as ESR uses a modulating field and a lock-in amplifier to detect resonance [11]. The integrated data can be seen in Figure 3 for when the orientation is 0° and in Figure 4 for a 30° orientation.

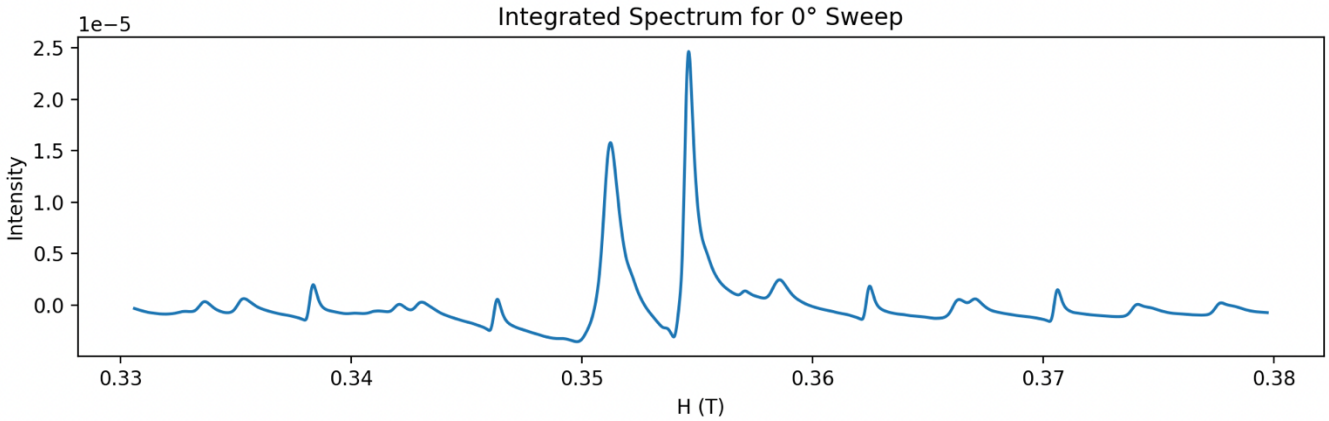


Figure 3: A plot of the integrated full sweep at an orientation of 0°.

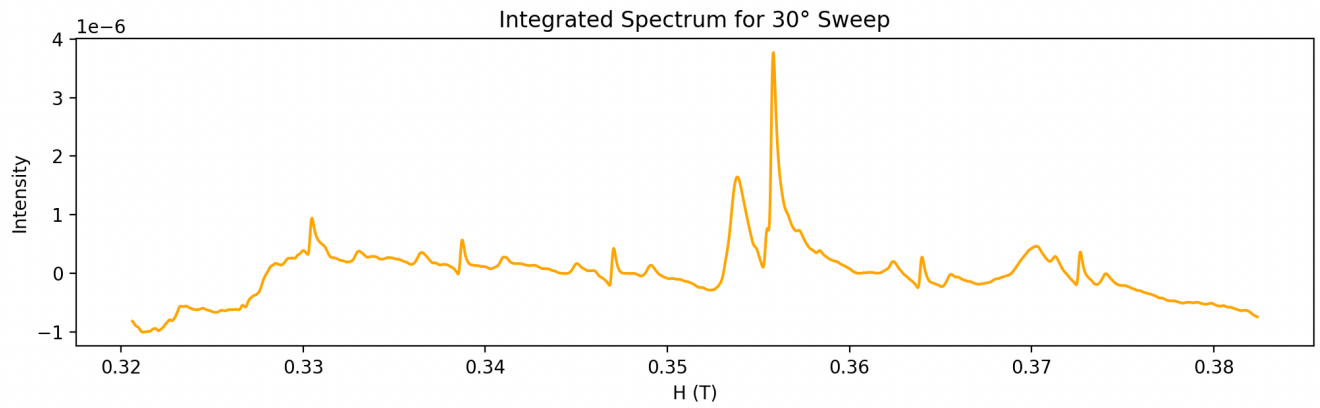


Figure 4: A plot of the integrated full sweep at an orientation of 30°.

The linear sweep should reveal the hyperfine interactions for the ion impurities in MgO. Fe^{3+} is expected to have 5 lines between the 6 levels where with the central line almost coinciding with the central line for DPPH. This makes sense since the g-factor for DPPH is g=2.0036 and for iron it is g=2.004. Iron would then have a H value of $H_{\mathrm{Fe}} = 0.3542\ T$ in comparison to the $H_{\mathrm{DPPH}} = 0.3546\ T$ for DPPH.

$$H_{\mathrm{Fe}} = \frac{\mathrm{h}\nu}{g\mu_B} = \frac{(6.626x10^{-34}\ Js)(9.945x10^{9}\ Hz)}{(2.004)\left(9.2741 \times 10^{-24}\ \frac{A}{m^2}\right)} = 0.3542\ T$$

The two middle peaks in Figure 3 and Figure 4 may both be attributed to Iron alongside the highest peak that is confidently identified as DPPH. The central peak for Chromium is calculated to be $H_{\mathrm{Cr}} = 0.3588\ T$ and $H_{\mathrm{Mn}} = 0.3550$ T.

$$H_{\mathrm{Cr}} = \frac{\mathrm{h}\nu}{g\mu_B} = \frac{(6.626x10^{-34}\ Js)(9.945x10^{9}\ Hz)}{(1.980)\left(9.2741 \times 10^{-24}\ \frac{A}{m^2}\right)} = 0.3588\ T$$

$$H_{\mathrm{Mn}} = \frac{\mathrm{h}\nu}{g\mu_B} = \frac{(6.626x10^{-34}\ Js)(9.945x10^{9}\ Hz)}{(2.0014)\left(9.2741 \times 10^{-24}\ \frac{A}{m^2}\right)} = 0.3550\ T$$

The magnetic field values for Fe^{3+}, Cr^{3+}, Mn^{2+} for each line in hyperfine splitting are calculated in the Appendix. The values for H for each ion impurity including hyperfine splitting can be compared to Figure 5 and Figure 6 where each peak is located and the corresponding H value is recorded in the legend. The comparison, using Figure 6, yielded a confirmation that peak 8 corresponds to both the central peak for Fe^{3+} and DPPH. The central peak for Cr^{3+} appears to correspond to peak 10 and Mn^{2+} with peak 9. There should be 6 equally spaced lines for Mn^{2+}. Peaks 3, 6, 11, and 12 may be contributed as being a part of the 6 lines for Mn^{2+} due to their equidistance and equal amplitude with peak 9.

5 Results and Discussion

This section discusses the results of the data collection from the linear sweeps at 2 different microwave orientations. The raw data for the derivative spectrum and the integrated spectrum are discussed alongside the theorizations on the location of the ion impurities.

5.1 Ion Impurities

The MgO crystal contains transition-metal impurities (Fe^{3+}, Cr^{3+}, Mn^{2+}) that each have ESR spectra that display hyperfine splitting. The H values for each energy level and resonance line were calculated in the Appendix. The highest peak that is clearly distinguishable is seen across all forementioned figures corresponds to the central peak for DPPH. This is peak 12 in Figure 5 and peak 8 in Figure 6. This is to be expected as DPPH should coincide with the most intense peak at $H_{\mathrm{DPPH}} = 0.3546\ T$ due to it having no fine or hyperfine interactions and is isotropic. A peak that is harder to detect is the Fe^{3+} central peak. This was also foreseeable as the g-factor for DPPH is g=2.0036 and g=2.004 for Iron. The central peak for Fe^{3+} could also be attributed to peak 12 in Figure 5 and peak 8 in Figure 6 and potentially the second highest peak, which is peak 11 in Figure 5 and peak 7 in Figure 6. This could be so that it is indistinguishable from the central line for DPPH and it accounted for the same value by shifting it slightly.

Mn^{2+} should have 6 equally spaced lines ranging from 1/2 to -1/2 . The central peak could correspond to peak 9 in Figure 6 and peak 13 in Figure 5, although the values slightly differ from the expected value of $H_{\mathrm{Mn}} = 0.3550\ \mathrm{T}$. The other 5 lines that are expected at $H_{\mathrm{Mn}} = 0.3767, 0.3563, 0.3507, 0.3569, 0.3680, 0.3593\ \mathrm{T}$ could include peaks 3, 6, 11, and 12 in Figure 6 given their equal spacing and similar height, however they differ from their expected values. This dissimilarity could be attributed to the changing of the orientation of the crystal and additional factors that could affect the value of the magnetic field.

Cr^{3+} appears to have a central at peak 10 in Figure 6 that corresponds to $H = 0.3585\ T$ which is close to the expected value of $H_{\mathrm{Cr}} = 0.3588\ T$. It should have 4 equally spaced resonance lines that are closely spaced together at values of $H_{\mathrm{Cr}} = 0.3563, 0.3580, 0.3598, 0.3615\ T$.

5.2 Sources of Error

Potential sources of error in this experiment could include incorrectly tuned sensitivity and time constant of the lock-in amplifier. Over the course of the experiment these parameters were continuously tuned, however due to the constant overloading and problems from the lock-in amplifier, these parameters could have been more optimized. The frequency that was also being regularly supplied was observed to be slightly fluctuating which would affect the value of the field and other values dependent upon the frequency. Additional noise could also have an

impact on the ESR spectra, such as any electromagnetic interference from laptops, etc. This could distort the signal and impact the data collection.

6 Conclusion

The purpose of this experiment was to gain a better understanding of the quantum physics that lays the foundation of Electron Spin Resonance and what can be learned from studying the ESR spectrum. A Varian Reflex Klystron alongside a calibration source of DPPH was used during the experiment to identify the ion impurities in MgO through analyzing its ESR spectrum. This was done by performing a linear sweep and observing and recording the magnetic field.

After locating the central DPPH peak and sweeping over an optimal range given the theoretical value of $H_{\mathrm{DPPH}} = 0.3546\,T$ from the Gaussmeter, we began sweeping for addition lines. A few peaks of interest became apparent after performing various sweeps with an appropriate time constant and sensitivity. From the integrated data, the peaks of interest could be compared to the expected magnetic field values for each ion impurity that were calculated. However, it was apparent that although some of the central peaks could be distinguished, not all of the resonance lines could be discerned. This could be attributed to not sufficient enough tuning, systematic errors, such as a varying frequency, and not sweeping a large enough range.

Appendix

A1 Plots with Located Peaks

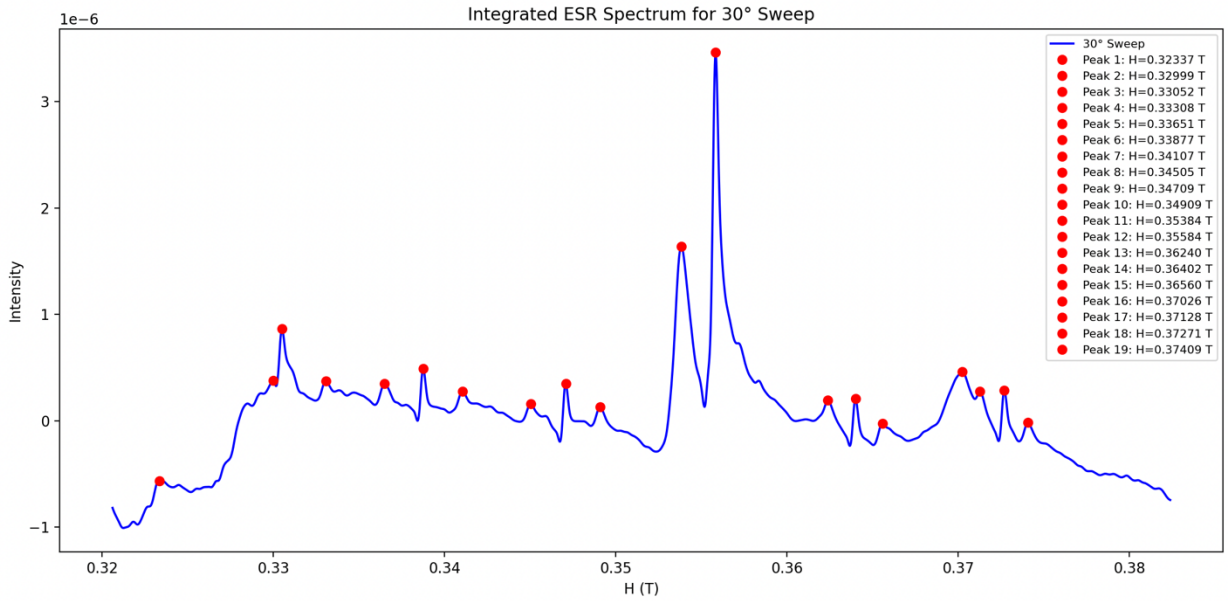


Figure 5: A plot of the integrated full sweep at an orientation of 30° with recorded peaks.

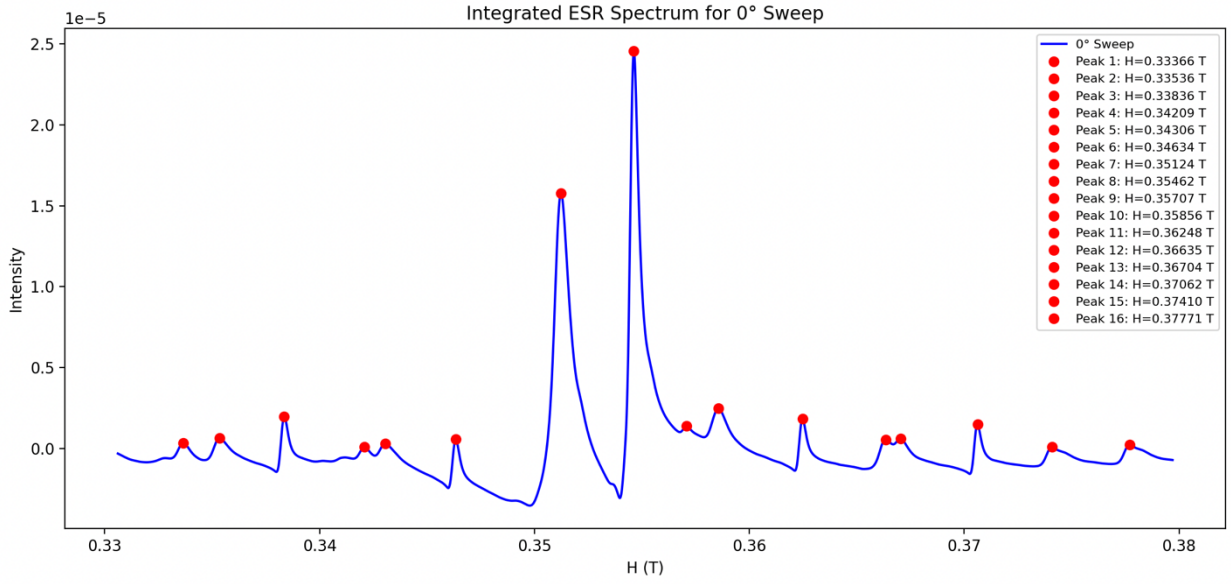


Figure 6: A plot of the integrated full sweep at an orientation of 0° with recorded peaks.

A2 Hyperfine Splitting Calculations

Iron

$$H_{\mathrm{Fe}} = \frac{\mathrm{h}\nu}{g\mu_B} = \frac{(6.626x10^{-34}\ Js)(9.945x10^{9}\ Hz)}{(2.004)\left(9.2741 \times 10^{-24}\ \frac{A}{m^2}\right)} = 0.3542\ T$$

$$H_{\mathrm{Fe}}\left(\mathrm{m_I} = \frac{1}{2}\right) = \frac{\mathrm{h}\nu}{g\mu_B} - \frac{\mathrm{Am_I}}{g\mu_B} = 0.3542 - \frac{1}{2} * 0.0005 = 0.3539\ T$$

$$H_{\mathrm{Fe}}\left(\mathrm{m_I} = -\frac{1}{2}\right) = \frac{\mathrm{h}\nu}{g\mu_B} - \frac{\mathrm{Am_I}}{g\mu_B} = 0.3542 + \frac{1}{2} * 0.0005 = 0.3544\ T$$

Chromium

$$H_{\mathrm{Cr}} = \frac{\mathrm{h}\nu}{g\mu_B} = \frac{(6.626x10^{-34}\ Js)(9.945x10^{9}\ Hz)}{(1.980)\left(9.2741 \times 10^{-24}\ \frac{A}{m^2}\right)} = 0.3588\ T$$

$$H_{\mathrm{Cr}}\left(\mathrm{m_I} = \frac{3}{2}\right) = \frac{\mathrm{h}\nu}{g\mu_B} - \frac{\mathrm{Am_I}}{g\mu_B} = 0.3588 - \frac{3}{2} * 1.7308\mathrm{x}10^{-3} = 0.3563\ T$$

$$H_{\mathrm{Cr}}\left(\mathrm{m_I} = \frac{1}{2}\right) = \frac{\mathrm{h}\nu}{g\mu_B} - \frac{\mathrm{Am_I}}{g\mu_B} = 0.3588 - \frac{1}{2} * 1.7308\mathrm{x}10^{-3} = 0.3580\ T$$

$$H_{\mathrm{Cr}}\left(\mathrm{m_I} = -\frac{1}{2}\right) = \frac{\mathrm{h}\nu}{g\mu_B} - \frac{\mathrm{Am_I}}{g\mu_B} = 0.3588 + \frac{3}{2} * 1.7308\mathrm{x}10^{-3} = 0.3598\ T$$

$$H_{\mathrm{Cr}}\left(\mathrm{m_I} = -\frac{3}{2}\right) = \frac{\mathrm{h}\nu}{g\mu_B} - \frac{\mathrm{Am_I}}{g\mu_B} = 0.3588 + \frac{3}{2} * 1.7308\mathrm{x}10^{-3} = 0.3615\ T$$

Manganese

$$H_{\mathrm{Mn}} = \frac{\mathrm{h}\nu}{g\mu_B} = \frac{(6.626x10^{-34}\ Js)(9.945x10^{9}\ Hz)}{(2.0014)\left(9.2741 \times 10^{-24}\ \frac{A}{m^2}\right)} = 0.3550\ T$$

$$H_{\mathrm{Mn}}\left(\mathrm{m_I} = \frac{5}{2}\right) = \frac{\mathrm{h}\nu}{g\mu_B} - \frac{\mathrm{Am_I}}{g\mu_B} = 0.3550 - \frac{5}{2} * -8.669\mathrm{x}10^{-3} = 0.3767\ T$$

$$H_{\mathrm{Mn}}\left(\mathrm{m_I} = \frac{3}{2}\right) = \frac{\mathrm{h}\nu}{g\mu_B} - \frac{\mathrm{Am_I}}{g\mu_B} = 0.3550 - \frac{3}{2} * -8.669\mathrm{x}10^{-3} = 0.3563\ T$$

$$H_{\mathrm{Mn}}\left(\mathrm{m_I} = \frac{1}{2}\right) = \frac{\mathrm{h}\nu}{g\mu_B} - \frac{\mathrm{Am_I}}{g\mu_B} = 0.3550 - \frac{5}{2} * -8.669\mathrm{x}10^{-3} = 0.3507\ T$$

$$H_{\mathrm{Mn}}\left(\mathrm{m_I}=-\frac{1}{2}\right)=\frac{\mathrm{h}\nu}{g\mu_B}-\frac{\mathrm{Am_I}}{g\mu_B}=0.3550+\frac{1}{2}*-8.669\mathrm{x}10^{-3}=0.3569\,T$$

$$H_{\mathrm{Mn}}\left(\mathrm{m_I}=-\frac{3}{2}\right)=\frac{\mathrm{h}\nu}{g\mu_B}-\frac{\mathrm{Am_I}}{g\mu_B}=0.3550+\frac{3}{2}*-8.669\mathrm{x}10^{-3}=0.3680\,T$$

$$H_{\mathrm{Mn}}\left(\mathrm{m_I}=-\frac{5}{2}\right)=\frac{\mathrm{h}\nu}{g\mu_B}-\frac{\mathrm{Am_I}}{g\mu_B}=0.3550+\frac{5}{2}*-8.669\mathrm{x}10^{-3}=0.3593\,T$$

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